Atherosclerosis and disc degeneration/low-back pain--a systematic review.

OBJECTIVES: Atherosclerosis can obstruct branching arteries of the abdominal aorta, including four paired lumbar arteries and the middle sacral artery that feed the lumbar spine. The diminished blood flow could result in various back problems. The aim of this systematic literature review was to assess associations between atherosclerosis and disc degeneration (DD) or low-back pain (LBP). DATA SOURCES: A systematic search of the Medline/PubMed database for all original articles on atherosclerosis and DD/LBP published until October 2008. The search was performed with the medical subject headings atherosclerosis, cardiovascular risk factor, or vascular disease and keywords "disc degeneration", "disc herniation", and "back pain" on the basis of MeSH tree and as a text search. In addition reference lists were studied and searched manually. Observational studies investigating the association of atherosclerosis or its risk factors and lumbar DD/LBP were selected. REVIEW METHODS: The following data were extracted: study characteristics, duration of follow-up, year of publication, findings of atherosclerosis/cardiovascular risk factors and DD/LBP. Disc herniation was regarded as a form of disc degeneration and cardiovascular risk factors were regarded as surrogate for atherosclerosis in epidemiological studies. RESULTS: One hundred and seventy-nine papers were identified. After exclusion of case reports, letters, editorials, papers not related to the lumbar spine, and animal studies, 25 papers were included. Post-mortem studies showed an association between atheromatous lesions in the aorta and DD, as well as between occluded lumbar arteries and life-time LBP. In clinical studies, aortic calcification was associated with LBP, and stenosis of lumbar arteries was associated with both DD and LBP. In epidemiological studies, smoking and high serum cholesterol levels were found to have the most consistent associations with DD and LBP. CONCLUSION: Aortic atherosclerosis and stenosis of the feeding arteries of the lumbar spine were associated with DD and LBP. Cardiovascular risk factors had weaker associations, being clearly apparent only in cohorts on elderly people or in large study samples. More prospective clinical studies are needed to further clarify the association of atherosclerosis and low-back disorders.